Chemistry • Electron transfer, sharing and intermolecular forces

Chemical Bonding

Official exercise • Complete solved answers • Original bonding diagrams

Chapter 3 • Complete official exercise • 10 MCQs + 6 short + 6 constructed + 6 descriptive + 2 investigative

What this chapter is about

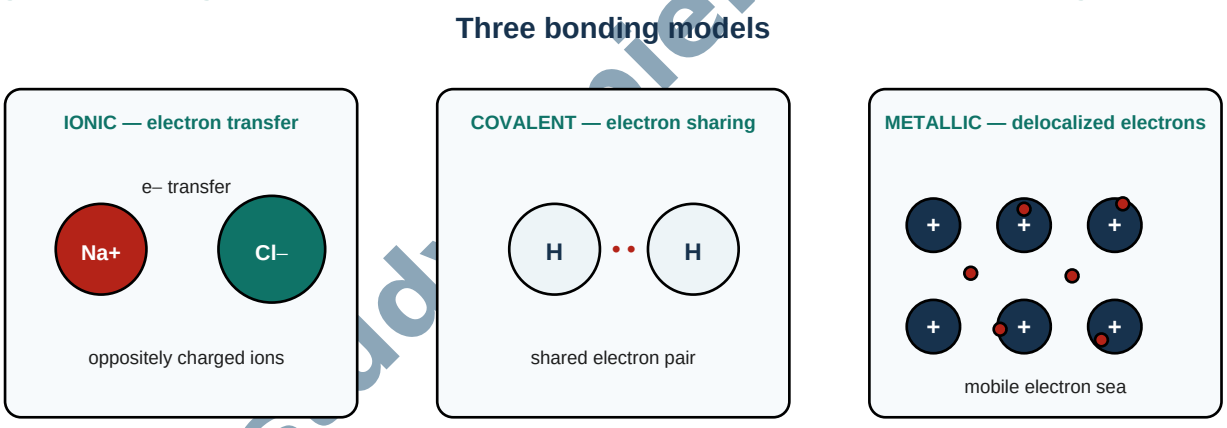
• **Chemical bonds** are attractive forces that hold atoms or ions together. Atoms bond to obtain more stable, lower-energy arrangements, often achieving a duplet or octet.

• **Ionic bonding** involves electron transfer and electrostatic attraction; **covalent bonding** involves shared pairs; a coordinate covalent pair is initially supplied by one atom.

• **Metallic bonding** is the attraction between metal cations and delocalized electrons. It explains conductivity, malleability and ductility.

• **Intermolecular forces** act between molecules. Dipole–dipole attractions, hydrogen bonding and London dispersion strongly influence melting points, boiling points and solubility.

Figure 1 — Original comparison of ionic, covalent and metallic bonding



Bond type explains structure, melting behaviour, solubility and electrical conductivity.

Exercise coverage: All five official sections are solved in book order. Formulae, option order and the ten-item MCQ sequence were checked against the textbook render.

A. Multiple Choice Questions

1. When molten copper and molten zinc are mixed together, they give rise to a new substance called brass. Predict what type of bond is formed between copper and zinc.

(a) coordinate covalent bond (b) ionic bond
(c) metallic bond (d) covalent bond

Answer: (c) metallic bond — Brass is a metallic alloy. Positive metal ions are held by delocalized electrons in a metallic lattice.

2. Which element is capable of forming all three types of bonds: covalent, coordinate covalent and ionic?

(a) carbon
(b) oxygen
(c) magnesium
(d) silicon

Answer: (b) oxygen — Oxygen shares electrons covalently, donates a lone pair in species such as H_3O^+, and forms O^{2-} in ionic oxides.

3. Why is H_2O a liquid while H_2S is a gas?

(a) because in water the atomic size of oxygen is smaller than that of sulphur
(b) because water is a polar compound and strong forces of attraction exist between its molecules
(c) because H_2O molecule is lighter than H_2S
(d) because water can easily freeze into ice

Answer: (b) because water is a polar compound and strong forces of attraction exist between its molecules — Water molecules form strong hydrogen bonds, whereas H_2S has much weaker intermolecular attractions and a much lower boiling point.

4. Which of the following bonds is expected to be the weakest?

(a) C–C
(b) Cl–Cl
(c) O–O
(d) F–F

Answer: (c) O–O — The O–O single bond is weakest because repulsion between lone pairs on adjacent small oxygen atoms weakens it (roughly 146 kJ mol^{-1}).

5. Which form of carbon is used as a lubricant?

(a) coal
(b) diamond
(c) graphite
(d) charcoal

Answer: (c) graphite — Weak attractions between graphite layers allow the layers to slide, giving graphite its lubricating action.

6. Keeping in view the intermolecular forces of attraction, indicate which compound has the highest boiling point.

(a) H_2O
(b) H_2S
(c) HF
(d) NH_3

Answer: (a) H_2O — Water has the highest listed boiling point (100°C) because each molecule can form an extensive three-dimensional hydrogen-bond network.

7. Which metal has the lowest melting point?

(a) Li
(b) Na
(c) K
(d) Rb

Answer: (d) Rb — Among these alkali metals, rubidium has the weakest metallic bonding and the lowest melting point (about 39°C).

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8. Which ionic compound has the highest melting point?

(a) NaCl (b) KCl
(c) LiCl (d) RbCl

Answer: (a) NaCl — NaCl has the highest melting point among the listed salts (about 801°C); the trend also reflects increasing polarization/covalent character in LiCl.

9. Which compound contains both covalent and ionic bonds?

(a) $MgCl_2$ (b) NH_4Cl
(c) CaO (d) PCl_3

Answer: (b) NH_4Cl — NH_4Cl contains covalent N–H bonds inside NH_4^+ and ionic attraction between NH_4^+ and Cl^-.

10. Which among the following has a double covalent bond?

(a) ethane (b) methane
(c) ethylene (d) acetylene

Answer: (c) ethylene — Ethylene (ethene), C_2H_4, contains a C=C double bond; ethane and methane have single bonds and acetylene has a triple bond.

B. Questions for Short Answers

3.1. What type of elements lose their outer electrons easily and what type of elements gain electrons easily?

Metals, especially groups 1 and 2, have few outer electrons and relatively low ionization energies, so they lose electrons and form cations. **Non-metals**, especially groups 16 and 17, have nearly complete outer shells and high electron affinity, so they gain electrons and form anions.

3.2. Why do lower molecular mass covalent compounds exist as gases or low-boiling liquids?

Their molecules are small and have relatively weak London dispersion or dipole attractions between them. Little thermal energy separates the molecules, so boiling points are low. Boiling overcomes **intermolecular** attractions; it does not normally break the strong covalent bonds inside each molecule.

3.3. Give one example of an element which exists as a crystalline solid and has covalent bonds between its atoms.

Diamond is a crystalline form of carbon in which each carbon atom forms four strong covalent bonds in a three-dimensional giant network. Graphite is another example, with covalently bonded hexagonal layers.

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3.4. Which property of metals makes them malleable and ductile?

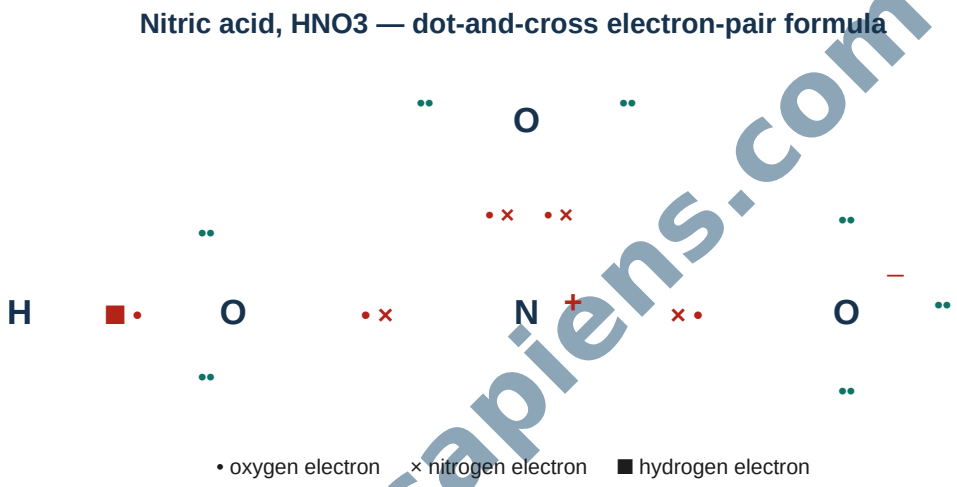
The **non-directional nature of metallic bonding** and mobile electron sea allow layers of metal cations to slide while remaining attracted to the surrounding electrons. The lattice can change shape without the bonds all breaking, so metals can be hammered into sheets (malleability) or drawn into wires (ductility).

3.5. Is a coordinate covalent bond a strong bond?

Yes. Once formed, a coordinate covalent bond is an ordinary shared-electron-pair bond and can be just as strong as another comparable covalent bond. Only the **origin** of the shared pair differs: initially both electrons come from the donor atom.

3.6. Write down the dot-and-cross formula of HNO_3.

Figure 2 — Original dot-and-cross Lewis formula; one resonance form



Each bonding pair is shared. One charged resonance form is shown; the two non-H N–O bonds resonate.

Alternative textbooks may omit formal charges; the total valence-electron count remains 24.

Check: H contributes 1 electron, N contributes 5 and three O atoms contribute 18, giving **24 valence electrons**. The displayed form has 5 bonding pairs and 7 lone pairs: 10 + 14 = 24 electrons.

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C. Constructed Response Questions

3.1. Why is HF a liquid while HCl is a gas?

HF molecules form strong **hydrogen bonds** because H is bonded to small, highly electronegative F. Much more energy is therefore needed to separate HF molecules; its boiling point is about 19.5°C, compared with about −85°C for HCl. HCl has dipole and London attractions but does not form a comparable hydrogen-bond network. Thus HF is liquid near the lower end of room-temperature conditions, whereas HCl is a gas.

3.2. Why are covalent compounds generally not soluble in water?

Many covalent compounds are non-polar and cannot form sufficiently strong attractions with polar water molecules to replace water–water hydrogen bonding and solute–solute attractions. They therefore remain separate. This is a trend, not an absolute rule: polar covalent substances such as sugar and ethanol dissolve through hydrogen bonding, and HCl dissolves while reacting/ionizing in water.

3.3. How do metals conduct heat?

At the hotter end, metal ions and free electrons gain kinetic energy. Delocalized electrons move rapidly through the lattice, collide with other electrons and ions, and transfer energy to cooler regions. Lattice vibrations also transmit energy, but mobile electrons make metals especially good thermal conductors.

3.4. How many oxides does nitrogen form? Write down the formulae of the oxides.

Nitrogen forms several oxides because it can show different oxidation states. Five principal simple oxides are commonly listed:

Oxide	Formula	Oxidation state of N
Dinitrogen monoxide (nitrous oxide)	N_2O	+1
Nitrogen monoxide (nitric oxide)	NO	+2
Dinitrogen trioxide	N_2O_3	+3
Nitrogen dioxide	NO_2	+4
Dinitrogen pentoxide	N_2O_5	+5

Nitrogen dioxide also exists in equilibrium with its dimer, **dinitrogen tetroxide, N_2O_4**: $2NO_2 \rightleftharpoons N_2O_4$. Thus N_2O_4 is often included when listing nitrogen–oxygen compounds.

3.5. What will happen if NaBr is treated with $AgNO_3$ in water?

A **pale-yellow precipitate of silver bromide** forms. Sodium nitrate remains dissolved.

$$NaBr(aq) + AgNO_3(aq) \rightarrow AgBr(s)\downarrow + NaNO_3(aq)$$

Net ionic: $Ag^+(aq) + Br^-(aq) \rightarrow AgBr(s)\downarrow$

This double-displacement reaction occurs because AgBr has very low solubility in water. Na^+ and NO_3^- are spectator ions.

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3.6. Why does iodine exist as a solid while Cl_2 exists as a gas?

I_2 molecules are much larger and more polarizable than Cl_2 molecules and contain more electrons. Consequently, I_2 has much stronger London dispersion forces and a higher melting point. The stronger intermolecular attractions keep iodine in the solid state at room temperature, while the weaker attractions between Cl_2 molecules allow chlorine to be gaseous.

D. Descriptive Questions

3.1. Explain the formation of an ionic bond and a covalent bond.

Ionic bond: One or more electrons transfer from a metal atom to a non-metal atom. The metal becomes a cation and the non-metal an anion; electrostatic attraction between the oppositely charged ions forms the ionic bond and a lattice develops.

Example: $Na \rightarrow Na^+ + e^-$; $Cl + e^- \rightarrow Cl^-$

$Na^+ + Cl^- \rightarrow$ NaCl lattice

Covalent bond: Two atoms share one or more electron pairs so that each atom counts the shared electrons towards its outer shell. One shared pair gives a single bond, two give a double bond and three give a triple bond.

Example: H• + •H → H:H or H–H

Feature	Ionic bond	Covalent bond
Electron process	Transfer	Mutual sharing
Usual participants	Metal and non-metal	Non-metal atoms
Particles produced	Cations and anions	Molecules or giant covalent networks
Main attraction	Opposite charges	Shared pair attracted to both nuclei

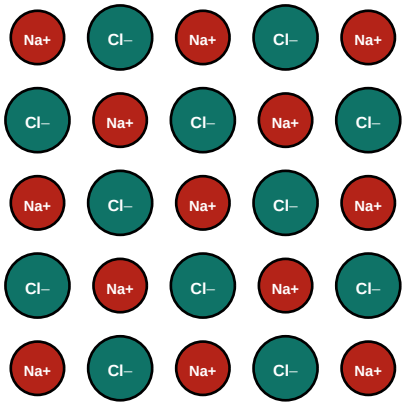
3.2. How do ions arrange themselves to form an NaCl crystal?

Na^+ and Cl^- ions arrange alternately in a regular three-dimensional lattice so that every ion is surrounded by ions of opposite charge. This maximizes electrostatic attraction and minimizes repulsion between like charges.

In the rock-salt structure, each Na^+ has six nearest Cl^- neighbours and each Cl^- has six nearest Na^+ neighbours, arranged octahedrally. The ions continue repeating in all directions; there is no single, separate NaCl molecule.

Figure 3 — Original two-dimensional slice of the three-dimensional NaCl lattice

Sodium chloride crystal lattice (2-D slice)



- Ions alternate in three dimensions
- Each Na+ has six Cl– neighbours
- Each Cl– has six Na+ neighbours
- Electrostatic attractions extend throughout the lattice
- **No separate NaCl molecules exist**

The diagram is only a slice: in the real 3-D rock-salt structure, coordination number = 6.

The formula NaCl is therefore an **empirical ratio** (1:1), not the formula of an isolated molecule.

3.3. Explain the properties of metals keeping in view the nature of metallic bond.

Metallic bonding is the attraction between a regular array of positive metal ions and delocalized electrons that move throughout the structure.

Property	Explanation from metallic bonding
Electrical conduction	Delocalized electrons drift towards the positive terminal when a potential difference is applied.
Thermal conduction	Mobile electrons and lattice vibrations transfer kinetic energy from hotter to cooler regions.
Malleability and ductility	Ion layers can slide while the non-directional electron sea continues to bind the structure.
Lustre	Free electrons absorb and re-emit visible light at the surface.
Generally high melting point	Strong attraction between cations and the electron sea requires considerable energy to disrupt.
Hardness and density	Strong bonding and close packing often produce hard, dense solids, although alkali metals are notable soft, low-density exceptions.

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3.4. Compare the properties of ionic and covalent compounds.

Property	Ionic compounds	Covalent compounds
Structure	Extended lattice of ions	Discrete molecules or giant covalent network
Usual state at room temperature	Crystalline solids	Small molecules often gases or liquids; larger molecules may be solids
Melting and boiling points	Usually high because strong electrostatic forces extend through lattice	Simple molecular substances often low; giant networks very high
Electrical conductivity	Do not conduct when solid; conduct when molten or aqueous because ions move	Usually non-conductors; graphite is an important exception
Water solubility	Many are water-soluble, but not all	Non-polar substances usually insoluble; polar ones may dissolve
Organic-solvent solubility	Usually low	Many non-polar molecular compounds dissolve
Examples	NaCl, MgO, $CaCO_3$	H_2O, CO_2, CH_4, SiO_2, diamond

Key distinction: melting a simple molecular covalent solid separates molecules; melting an ionic solid or giant covalent structure requires overcoming attractions or bonds extending throughout the lattice.

3.5. How will you explain the electrical conductivity of graphite crystals?

In graphite, each carbon atom forms three covalent bonds in a hexagonal layer. Four outer electrons are available per carbon, but only three are used in these sigma bonds.

The remaining electron from each carbon becomes part of a **delocalized π-electron system** over the layer. These electrons can move in response to a potential difference, so graphite conducts electricity, especially parallel to its layers.

Diamond has no comparable delocalized electrons because all four outer electrons of each carbon are localized in four covalent bonds; it is therefore an electrical insulator.

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3.6. Why are metals usually hard and heavy?

Many metals have strong metallic bonding and closely packed atoms, so their lattices resist deformation and contain considerable mass per unit volume. Strong cation–electron attraction contributes hardness and high melting points, while close packing contributes high density. These are general trends rather than universal rules: sodium and potassium are soft and have low densities, whereas osmium is extremely dense.

E. Investigative Questions

3.1. The formula of $AlCl_3$ in vapour phase is Al_2Cl_6, which means it exists as a dimer. Explain the bonding between its two molecules.

An Al atom in $AlCl_3$ has only six electrons in its valence shell and is electron-deficient. Chlorine atoms have lone pairs. In the dimer, a chlorine atom bonded to one aluminium donates a lone pair to the other aluminium, forming a **coordinate covalent (dative) bond**.

Two chlorine atoms act as bridges, producing a four-membered Al–Cl–Al–Cl ring. The other four chlorines are terminal, two on each aluminium. Each aluminium therefore has four surrounding chlorine atoms and achieves a more stable octet.

$$2\ AlCl_3 \rightarrow Al_2Cl_6$$

Dimer: $Cl_2Al(\mu\text{-}Cl)_2AlCl_2$, where μ-Cl means bridging chlorine

The bridging links may be represented by arrows Cl → Al to show the donor and acceptor. After formation, each coordinate link is a shared-pair covalent bond.

3.2. Explain the structure of sand (SiO_2).

Sand is mainly silicon dioxide, a **giant covalent network solid**. Each silicon atom forms four single covalent bonds to oxygen atoms arranged tetrahedrally. Each oxygen atom bridges two silicon atoms.

SiO_4 tetrahedra share corners; overall ratio = 1 Si : 2 O

No separate SiO_2 molecule exists in quartz; SiO_2 is an empirical formula. Strong Si–O bonds extend throughout the network, explaining its hardness, high melting point, insolubility in water and lack of electrical conductivity.

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Points to Remember

Ionic bond: electron transfer followed by electrostatic attraction between ions.

Covalent bond: mutual sharing of electron pairs; single, double and triple bonds contain one, two and three shared pairs.

Coordinate covalent bond: both electrons of the shared pair are initially donated by one atom; after formation it behaves as a covalent bond.

Metallic bond: attraction between metal cations and delocalized electrons; explains conductivity and deformability.

Hydrogen bonding occurs strongly in molecules containing H bonded to N, O or F and raises boiling points.

London dispersion increases with molecular size and polarizability; it explains why I_2 is solid while Cl_2 is gaseous.

Solid ionic compounds do not conduct because ions are fixed; molten and aqueous ionic compounds conduct because ions move.

Graphite conducts through delocalized electrons within layers; diamond does not conduct because its electrons are localized.

MCQ key at a glance

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	c	2	b	3	b	4	c	5	c
6	a	7	d	8	a	9	b	10	c

How to attempt this chapter in the paper

• State whether the question concerns bonds **within** particles or attractions **between** particles before explaining a physical property.
• For conductivity, always name the mobile charge carrier and the conditions: electrons in metals/graphite, ions in molten/aqueous electrolytes.
• For Lewis structures, count total valence electrons, complete octets where applicable and check formal charges.
• For dative bonds, use an arrow from lone-pair donor to acceptor, but remember that the completed bond is a shared-pair bond.

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